$\boxed{\text{properties of supremum and infimum}}$

1. additive property

$A, B \subset \mathbb{R}$, nonempty, $C = \{a+b \mid a \in A, b \in B\}$

(a) $\sup A, \sup B \ \exists \Rightarrow \sup C \ \exists$, $\sup C = \sup A + \sup B$

(b) $\inf A, \inf B \ \exists \Rightarrow \inf C \exists, \inf C = \inf A + \inf B$

proof: (a) ① $\forall c \in C, \exists a \in A, b \in B, c = a+b$

$\therefore c \leq \sup A + \sup B$

$\therefore \sup C \exists$, $\sup C \leq \sup A + \sup B$

② $\forall n \in \mathbb{Z}^+$, $\exists a \in A, b \in B$,

s.t., $a > \sup A - \frac{1}{n}$, $b > \sup B - \frac{1}{n}$

$\therefore \sup C \geqslant a + b > \sup A + \sup B - \frac{2}{n}$

$\therefore \sup C = \sup A + \sup B$.

(b) The proof is similar.

2. $A \subset \mathbb{R}$, nonempty, $C = \{ka \mid a \in A\}$

(a) $\sup A \ \exists \Rightarrow \begin{cases} \sup C \exists, \sup C = k\sup A, k>0 \\ \inf C \exists, \inf C = k\sup A, k<0 \end{cases}$

(b) $\inf A \exists \Rightarrow \begin{cases} \inf C \exists, \inf C = k\inf A, k>0 \\ \sup C \exists, \sup C = k\inf A, k<0 \end{cases}$

proof: (a) if $k>0$

① $\forall c \in C, \exists a \in A, c = ka$

$\therefore c \leq k\sup A$

$\therefore \sup C \exists$, $\sup C \leq k\sup A$

② $\forall n \in \mathbb{Z}^+$, $\exists a \in A$, s.t., $a > \sup A - \frac{1}{n}$

$\therefore \sup C \geqslant ka > k\sup A - \frac{k}{n}$

$\therefore \sup C = k\sup A$.

if $k<0$, similar.

(b) The proof is similar.

$\blacksquare$

3. $S, T \subset \mathbb{R}$, nonempty,

$s \leqslant t, \forall s \in S, t \in T$.

$\Rightarrow \sup S, \inf T \ \exists, \sup S \leq \inf T$

proof: $\because s \leq t$, $\forall s \in S, t \in T$

$\therefore \sup S \exists$, $\sup S \leq t, \forall t \in T$

$\therefore \inf T \exists$, $\sup S \leq \inf T$.

4. $S, T \subset \mathbb{R}$, nonempty,

$S \subseteq T$, $\sup T, \inf T \ \exists$.

$\Rightarrow \sup S, \inf S \ \exists$,

$\sup S \leq \sup T$, $\inf S \geqslant \inf T$

proof: $\because s \leq \sup T$, $\forall s \in S$

$\therefore \sup S \exists$, $\sup S \leq \sup T$.

$\because s \geqslant \inf T$, $\forall s \in S$

$\therefore \inf S \exists$, $\inf S \geqslant \inf T$.

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